



E-13, UPSIDC Site-IV, Kasna Road, Greater Noida - 201308, India
Phone: +91.120.4296878 | Website: www.khuranaandkhurana.com www.iiprd.com
Email: info@khuranaandkhurana.com | iiprd@iiprd.com

Indian Offices: Noida | New Delhi | Bangalore | Mumbai | Pune | Mumbai | Hyderabad | Jalandhar | Chennai
International Offices: US | Bangladesh | Sri Lanka | Malaysia | Vietnam | Myanmar | Nepal | Malaysia

INVENTIVE DISCLOSURE - CONFIDENTIAL

1. **Proposed Title of the Invention**

EIDEN FIST

2. **Proposed Abstract of the Invention**

An EIDEN FIST is a wearable technology device that includes sensors, micro controllers, and other electronic components to enable a range of functions such as tracking movement, measuring biometric data, and controlling other devices. Here are some potential modules that could be used in it. Purpose of an EIDEN FIST is to provide a more natural and intuitive way for users to interact with digital systems and devices, whether for entertainment, health, or work-related purposes.

Some common **applications** of smart gloves include:

Virtual reality: - EIDEN FIST can be used to provide more immersive and intuitive experiences in virtual reality. The sensors in the gloves can track hand and finger movements, allowing users to control virtual objects and characters with their hands.

Health and rehabilitation: - EIDEN FIST can be used in physical therapy and rehabilitation to help patients regain strength and dexterity in their hands and fingers after an injury or surgery. The sensors in the gloves provide real-time feedback on the patient's movements and progress.

Industrial and manufacturing:- EIDEN FIST can be used in industrial and manufacturing settings to improve worker safety and productivity. The gloves can monitor workers' hand movements and provide alerts if they are performing tasks incorrectly or are at risk of injury.

3. Key Words:

Wearable technology gloves, Smart wearable gloves

4. Background of the Invention:

The market for **EIDEN FIST** is diverse and spans across multiple industries, including gaming, virtual reality, health and rehabilitation, industrial manufacturing, and more. Here's an overview of the market for smart gloves, including the target audience, competitors, and potential sales:

1. Gaming and virtual reality- The gaming and virtual reality market is a significant target audience for smart gloves. Smart gloves can enhance the user's immersion and provide a more natural and intuitive way to interact with virtual environments. Competitors in this market include companies such as HaptX, Manus VR, and Teslasuit. The potential sales for this market are projected to reach **\$29.5 billion** by 2025.

2. Health and rehabilitation- Smart gloves can be used in physical therapy and rehabilitation to help patients regain strength and dexterity in their hands and fingers after an injury or surgery. Competitors in this market include companies such as Neofect and Kinestica. The potential sales for this market are projected to reach **\$6.7 billion** by 2025.

3. Industrial and manufacturing- Smart gloves can be used in industrial and manufacturing settings to improve worker safety and productivity. Competitors in this market include companies such as ProGlove and Eonite Perception. The potential sales for this market are projected to reach **\$2.2 billion** by 2025.

4. Military and defense- Smart gloves can be used in military and defense applications to provide soldiers with enhanced control and communication capabilities in the field. Competitors

in this market include companies such as HaptX and BeBop Sensors. The potential sales for this market are projected to reach **\$3.3 billion** by 2025.

Overall, the market for smart gloves is projected to experience significant growth in the coming years, with potential sales reaching over \$40 billion by 2025. The target audience for smart gloves includes gamers, virtual reality enthusiasts, patients in rehabilitation, industrial and manufacturing workers, and military personnel.

5. What problems does the invention address and how your Invention is able to overcome the limitations/ problems of the existing technologies?

EIDEN FIST can provide several benefits and advantages in a wide range of applications, including gaming, virtual reality, health and rehabilitation, industrial manufacturing, and more. Here are some of the **reasons** why we need it.

1. Natural and intuitive interaction- EIDEN FIST can provide a more natural and intuitive way for users to interact with digital devices and systems. Instead of using a keyboard or mouse, users can control devices and manipulate data using their hands and fingers.

2. Increased accuracy and precision- EIDEN FIST can enhance the accuracy and precision of user interactions with digital systems, allowing for greater control and fine-tuned movements.

3. Improved productivity- EIDEN FIST can streamline workflows and reduce the time and effort required to perform certain tasks, improving efficiency and productivity.

4. Enhanced safety- EIDEN FIST can improve worker safety in hazardous environments by providing real-time feedback and alerts to workers when they are at risk of injury or are performing tasks incorrectly.

5. Accessibility- EIDEN FIST can provide an alternative way for people with disabilities or mobility impairments to interact with digital devices and systems, improving their access to technology and enabling greater independence.

EIDEN FIST can provide a more natural, efficient, and engaging way for users to interact with digital devices and systems, improving productivity, safety, accessibility, and the overall

userexperience.

6. Detailed Explanation of the Invention along with working examples. Kindly provide an elaborated description of each and every aspect of the invention (product and/or process) in great detail.

1) ARDUINO NANO

Microcontroller (Arduino NANO) is a small, complete, and breadboard-friendly board based on ATmega328. It is the heart of the Smart Glove. It receives analog input from flexsensors and sends it to the buzzer accordingly. Atmega328 is a microcontroller that is widely used in Arduino boards; Arduino Nano and Uno both have Atmega328, which provides numerous features. Nano is a widely acceptable microcontroller board because of its small size and flexibility.

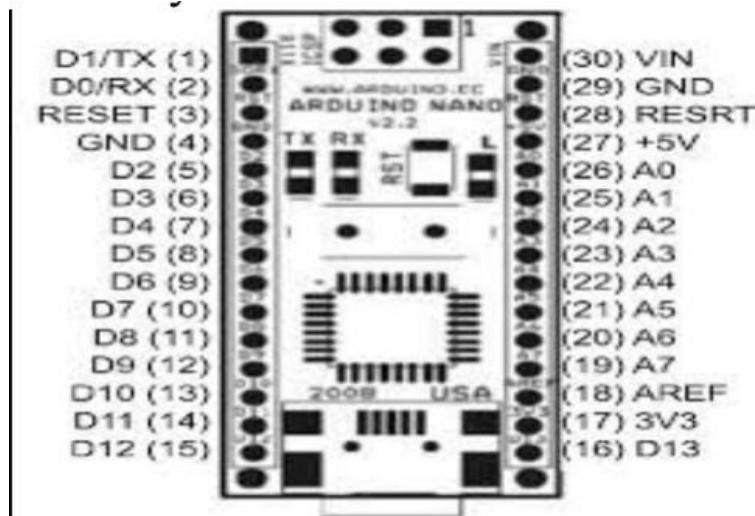


FIG:1 - Arduino Nano(ATmega328)

II) FLEX SENSOR

A flex sensor is a type of sensor that changes its resistance as it is bent. It typically consists of a flexible substrate, such as a thin strip of plastic or rubber, with conductive material, such as carbon or metal, applied to its surface. When the flex sensor is bent, the distance between the conductive materials on the surface changes, which alters the resistance of the sensor.

These sensors detect changes in the flexion and extension of fingers and can be used to measure finger movement or gesture recognition. Flex sensors are often used in smart gloves to detect the movement of the wearer's fingers. Smart gloves are used in a variety of applications, such as virtual and augmented reality, gaming, and medical rehabilitation. In a smart glove, flex

sensors are placed on the fingers. As the wearer moves their fingers, the flex sensor detects the degree of bend in the finger and transmits that information to an Arduino Nano. Overall, the usage of flex sensors in smart gloves allows for more natural and intuitive control of devices and provides a more immersive and interactive experience for the wearer.



FIG: 2- Flex Sensor (2.2 inch Bend Sensor)

III) SWITCH

Switches are used in smart gloves for a variety of reasons. Smart gloves often incorporate sensors, such as touch or pressure sensors, to detect gestures or movements made by the user's fingers or hand. Switches can be used to provide a way for the user to interact with the device and activate certain functions, such as turning on/off the gloves or activating a particular feature. Switches are also used to help conserve battery life in smart gloves. When the user is not actively using the gloves, the switches can be set to turn off certain functions or put the gloves into a low-power mode to extend the battery life.

In some cases, switches may also be used as a safety feature in smart gloves. For example, if the gloves are being used in a hazardous environment, a switch can be used to immediately shut off the gloves in case of an emergency.

Overall, switches are an important component of smart gloves as they provide a convenient way for the user to interact with the device and enable certain features while also helping to conserve battery life and ensure user safety.



FIG 3: - Switch (2Pin SPST On-O-On Rocker)

IV) LEDS USED IN AN OXIMETER

Pulse oximetry is a non-invasive method for determining hemoglobin (Hb) saturation through

oxygen in the blood and heart frequency. It uses the fact that oxygen-rich and oxygen-poor hemoglobin absorb red and infrared light differently. To measure the oxygen content reliably, LEDs with high performance, constant quality, and low scattering are needed as light sources.

The right components are essential to reliably determine the oxygen content of the blood. The LEDs and photodetectors that are used are particularly important. The red LED is characterized by an exact wavelength of 652 nm and high efficiency. With a current of $I_F=20\text{mA}$, a typical luminous intensity of 130 mcd is achieved. In combination with a good phototransistor, this results in a high signal-to-noise ratio. To obtain a good signal-to-noise ratio with the IR LED as well, it should have about twice the luminous intensity of the red LED in this application. The LED has a very high efficiency and high radiation intensity. The phototransistor is recommended as a light sensor. Although the component has the highest light sensitivity at a wavelength of 940 nm, it also detects light signals with a wavelength of 660 nm. Therefore, only one single phototransistor is needed for the measurement.

Oximeters take advantage of the fact that red and infrared light is absorbed differently by oxygen-rich and oxygen-poor blood: HbO_2 absorbs more infrared light and less red light than Hb. Due to the higher reflection of red light, it appears bright red. By contrast, Hb absorbs more red light and thus appears darker.

Transmissive pulse oximetry usually uses a red LED in the range of about 650 nm and an IR LED in the range of about 900 nm. Both LEDs screen a fingertip and a phototransistor on the other side captures the relative amount of absorbed red and IR light. A microcontroller then determines the proportion of Hb and HbO_2 and calculates the oxygen content from the measured parameters. Only the arterial blood is of interest. With pulse oximetry, it is possible to detect the oxygen content only in arterial blood. This is based on the fact that the amount of absorbed red and infrared light varies with every heartbeat due to the pulsating changes in blood volume. By contrast, veins, bones, and tissue show relatively constant absorption characteristics.



FIG: 4 -LEDs used in an oximeter (5mm LED)

V) BATTERY USED IN EIDEN FIST AND OXIMETER

Eden fist is a type of wearable technology that incorporates sensors and other electronic components to provide various functions such as tracking hand movements, monitoring vital signs, or controlling other devices.

In it, a battery is typically used to power the electronic components such as sensors, processors, and communication modules that are embedded in the glove. The battery provides a portable and self-contained power source that enables the smart glove to function independently without needing to be tethered to an external power source.

The type of battery used in a smart glove and oximeter can vary depending on the specific application and design requirements of the device. Common types of batteries used in smart gloves include lithium-ion (Li-ion) and lithium polymer (LiPo) batteries, which are lightweight, compact, and rechargeable. These types of batteries provide a high energy density and can deliver the necessary power to run the smart glove's electronic components for an extended period.

The battery is typically located in a compartment within the smart glove or in a separate battery pack that can be attached to the glove. Depending on the design of the smart glove, the battery may be replaceable or non-replaceable, and the charging method may be wired or wireless.



FIG: 5 - Battery used in EIDEN FIST and Oximeter (9V Battery (Hi-Waote))

VI) RESET BUTTON IN ARDUINO USED IN EIDEN FIST

The reset button on an Arduino board is used to restart the microcontroller, which is the brain of the board. When the reset button is pressed, the microcontroller is momentarily reset and the program running on the board starts executing from the beginning.

Glove - To initiate pairing mode, the reset button can be used to reset the microcontroller and put the Arduino into a special mode where it is ready to connect to other devices. Another possible use of the reset button in smart gloves is to reset the sensors or other components used in the glove. Smart gloves can use various sensors, such as flex sensors to detect the movements of the wearer's hands and fingers. If these sensors are not working correctly or are producing unexpected results, the reset button can be used to reset the microcontroller and reinitialize the sensors.

In addition, the reset button can also be used in the development and testing of smart gloves. When developing software for the glove, the reset button can be used to quickly restart the program and test changes made to the code. It can also be used to recover from errors or unexpected behavior during testing.

Overall, the reset button on an Arduino board can be a useful feature in smart gloves, allowing for easy and quick resetting of the microcontroller and the components connected to it.

Oximeter- The reset button is typically used to force the MCU to start executing code from the beginning as if the board had just been powered on. This can be useful in several situations, such

as when you're testing new code and want to start from a known state, or if the board is behaving unexpectedly and you want to restart it to try to resolve the issue.

In the context of an oximeter project, the reset button might be used to reset the MCU and the oximeter sensors to a known state before taking a new reading, or to start a new measurement cycle after the previous one has been completed.

Overall, the reset button on an Arduino board is an important feature that helps to ensure the reliable operation of the board and makes it easier to develop and debug programs for the board.

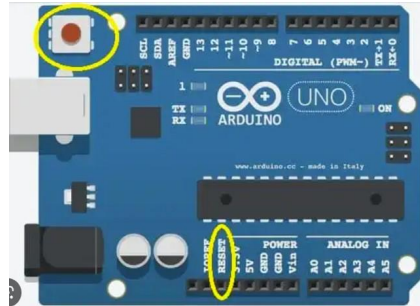


FIG: 6 - Reset Button in Arduino used in EIDEN FIST (Arduino ProMini 5V)

7. **Kindly attach drawings, reports, papers, charts or other materials that may aid in your description.**

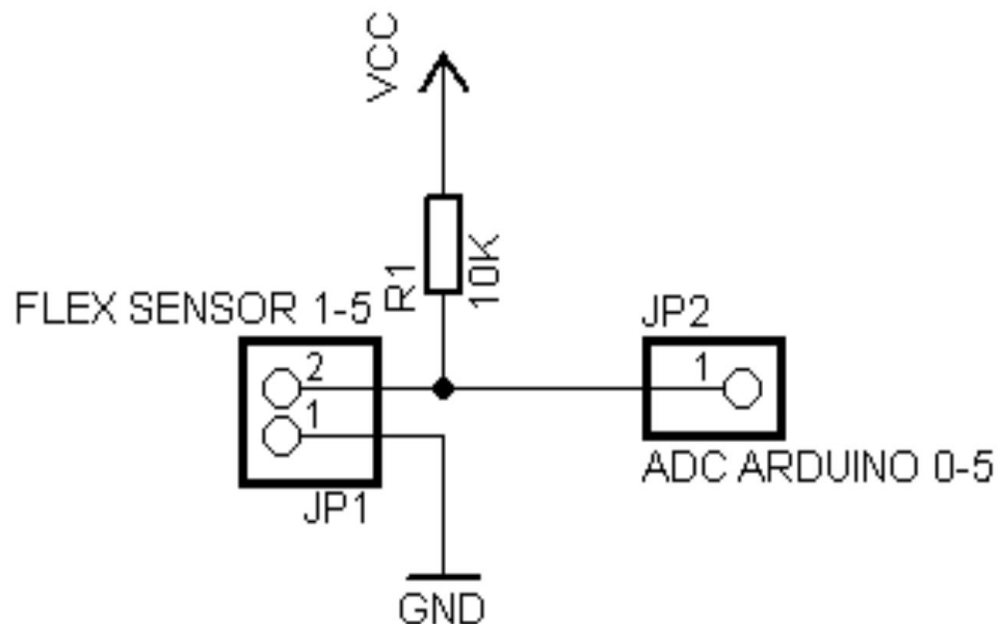


Fig:7 - Circuit diagram EIDEN FIST

8. What are the aspects of your disclosure that you want to claim/monopolize?

Proposed Claims:

Presenting the smart glove "EIDEN FIST", is a wearable technology device that includes sensors, microcontrollers, and other electronic components to enable a range of functions such as tracking movement, measuring biometric data, and controlling another device. EIDEN FIST can provide an alternative way for people with disabilities or mobility impairments to interact with digital devices and systems, improving their access to technology and enabling greater independence.

9. Have you conducted novelty/inventiveness search for your invention? If yes, what are the databases /references used by you? What are the search results?

There have been several recent inventions and developments in the field of smart gloves, which aim to provide various benefits and functionalities to users. Some notable examples include:

1. **Haptic feedback gloves:** These gloves incorporate small motors or vibration sensors to provide users with a tactile sensation or feedback. They can be used in virtual reality applications to simulate touch or in rehabilitation settings to help patients regain hand dexterity.
2. **Gesture recognition gloves:** These gloves use sensors to detect hand movements and gestures, allowing users to control devices or interact with virtual environments using hand gestures.
3. **Biometric monitoring gloves:** These gloves feature sensors that can monitor vital signs, such as heart rate and blood pressure, and provide real-time feedback to healthcare professionals or patients themselves.
4. **Smart gloves for sign language translation:** These gloves use sensors and machine learning algorithms to recognize and translate sign language into spoken language, making it easier for deaf or hard-of-hearing individuals to communicate with others.
5. **Smart gloves for sports training:** These gloves incorporate sensors to measure hand and wrist movements during sports training, providing athletes with real-time feedback on technique and performance.

10. Do you feel that a person of “average” skill (not-extraordinary skill) in your area of technology would have arrived at your invention with existing knowledge in public domain? If no, what could be the reasons for the same?

In many cases, inventions and discoveries are the result of incremental progress built on top of existing knowledge, and may not require exceptional skill or talent. However, some breakthroughs may require a unique combination of skills, insights, and expertise that are not readily available in the public domain.

Furthermore, there may be other factors that could affect a person's ability to arrive at an invention, such as access to resources, funding, and collaborations with experts in the field. The intellectual property landscape, including existing patents and licensing agreements, can also play a role in shaping the feasibility and profitability of a new invention.

Overall, the likelihood of an individual arriving at a particular invention or discovery depends on a complex interplay of factors, and may vary widely depending on the specific context and circumstances involved.

11. Kindly provide broad workable ranges for all the parameters involved in your invention.

Smart gloves can have a variety of parameters depending on their intended use and the technology they employ. However, some common parameters and their workable ranges are

1. SENSOR TYPE: Smart gloves can use a range of sensors, including flex sensors, accelerometer, gyroscope, pressure sensors, and touch sensors. The workable ranges of these sensors will depend on their specifications and the intended use. For example, a flex sensor may have a range of 45 to 90 degrees of bending, while an accelerometer may have a range of +/- 2g or +/- 16g.

2. Connectivity: Smart gloves may use different types of connectivity options, such as Bluetooth, Wi-Fi, or NFC. The range of Bluetooth and Wi-Fi will depend on the version and the device used, but typically the range is around 10-30 meters. NFC, on the other hand, has a range of a few centimeters.

3. BATTERY CAPACITY: The battery capacity will depend on the intended use of the smart gloves, but typically ranges from 100 mAh to 1000 mAh.

4. Battery life: The battery life will depend on the battery capacity and the power consumption of the sensors and other components. The battery life can range from a few hours to several days.

5. Operating temperature: The operating temperature of smart gloves will depend on the materials used and the electronics inside. Typically, the operating temperature ranges from -10°C to 40°C.

6. Storage temperature: The storage temperature of smart gloves will depend on the materials used and the electronics inside. Typically, the storage temperature ranges from -20°C to 60°C.

7. Waterproofing: The level of waterproofing will depend on the intended use of the smart gloves. For example, if the gloves are designed for use in wet environments, they may be waterproof to a depth of a few meters. If they are designed for general use, they may only be resistant to splashes and light rain.

12. References (if any)

Here are some references for EIDEN FIST:

1. "Smart gloves for virtual and remote interactions" by M. Gilliot, et al. in VirtualReality, 2021.

2. "A Review of Smart Gloves for Virtual and Augmented Reality" by Y. Yu and X. Wang in IEEE Access, 2019.

3. "Smart Gloves: A Comprehensive Review on Wearable Technology" by N. Shahid and A. Gani in Journal of Ambient Intelligence and Humanized Computing, 2020.

4. Hi-Fun Bluetooth Gloves: These gloves have a built-in microphone and speaker, allowing you to make phone calls and listen to music without taking off your gloves.

5. Agloves Touchscreen Gloves: These gloves have conductive fingertips that allow you to use your smartphone or tablet without taking them off.

6. Quell Relief Smart Wearable: These gloves are designed to provide pain relief for people with arthritis or other hand pain conditions. They use transcutaneous electrical nerve stimulation (TENS) technology to stimulate the nerves in the hands and reduce pain.

7. ProGlove: These smart gloves are designed for industrial use and feature built-in sensors that can track worker productivity, monitor inventory, and improve safety.

8. Vuzix Blade AR Smart Glasses with Gesture Control: These gloves work in conjunction with the Vuzix Blade AR smart glasses to allow users to control the glasses using hand gestures.

13. Inventors Details (Full Names, Nationality and Addresses)

1) Full name of the inventor: Dr. E. Umamaheswari

Designation: Professor-Grade-1

Emp ID : 50378

School: Centre for Cyber Physical Systems

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Intercom / Mobile No: 04439931194

E-mail: umamaheswari.e@vit.ac.in

2) Full name of the inventor: Dr. A. Karmel

Designation: Associate Professor Senior

Emp ID : 50435

School: SCOPE

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Intercom / Mobile No: +91 98405 07512

E-mail: karmel.a@vit.ac.in

3) Full name of the inventor: Dr. Kanchana Devi V

Designation : Associate Professor Grade 2

Emp ID: 50311

School: School of Computer Science and Engineering

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Mobile No.: +91 95006 42843

E-mail: kanchanadevi@vit.ac.in

4) Full name of the inventor: Dr. A. David Maxim Gururaj

Designation: Associate Professor Grade 2

Emp. ID: 50598

School: School of Advanced Sciences

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Mobile No.: +94435 08589

E-mail: davidmaxim.gururaj@vit.ac.in

5) Full name of the inventor: Adwaita Jha

Reg No: 21MIS1186

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

6) Full name of the inventor: Raj Koyani

Reg. No.: 21MIS1017

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

7) Full name of the inventor: Vaishnavi Shukla

Reg. No.: 21MIS1100

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

8) Full name of the inventor: Mr. Yogeshwar Ganesh

Reg. No: 19MIS1151

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam

road, Kancheepuram district, Tamilnadu 600127

14. Applicant Details (Full Names, Nationality and Addresses)

1) Full name of the inventor: Dr. E. Umamaheswari

Nationality: Indian

Designation: Professor

Emp ID : 50378

School: Centre for Cyber Physical Systems

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Intercom / Mobile No: 04439931194

E-mail: umamaheswari.e@vit.ac.in

2) Full name of the inventor: Dr. A. Karmel

Nationality: Indian

Designation: Associate Professor Senior

Emp ID : 50435

School: School of Computer Science and Engineering

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Intercom / Mobile No: +91 98405 07512

E-mail: karmel.a@vit.ac.in

3) Full name of the inventor: Dr. Kanchana Devi V

Nationality: Indian

Designation : Associate Professor Grade 2

Emp ID: 50311

School: School of Computer Science and Engineering

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

Mobile No.: +91 95006 42843

E-mail: kanchanadevi@vit.ac.in

4) Full name of the inventor: Dr.A.David Maxim Gururaj

Designation: Associate Professor Grade 2

Emp. ID: 50598

School: School of Advanced Sciences

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127 Mobile No.: +94435 08589

E-mail: davidmaxim.gururaj@vit.ac.in

5) Full name of the inventor: Adwaita Jha

Reg No: 21MIS1186

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

6) Full name of the inventor: Raj Koyani

Reg. No.: 21MIS1017

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

7) Full name of the inventor: Vaishnavi Shukla

Reg. No.: 21MIS1100

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam road, Kancheepuram district, Tamilnadu 600127

9) Full name of the inventor: Mr. Yogeshwar Ganesh

Reg. No: 19MIS1151

Nationality: Indian

Office Address: Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam

road, Kancheepuram district, Tamilnadu 600127

15. Any additional notes or remarks.

Smart gloves are a type of wearable technology that incorporate sensors, haptic feedback mechanisms, and other advanced technologies to enable users to interact with digital or physical environments in new and innovative ways. Here are some key points of smart gloves.

16. For Life Sciences related inventions:

- i. Provide source and geographical origin of the biological material/resource (for e.g. plants, animals, micro-organisms, their parts / genetic material and by-products with actual or potential use or value)**

No. Neither the source nor the geographical origin of the biological material/resource is used.

- ii. Please note, if the biological material used in the invention is from India, then an application to seek approval of the National Biodiversity Authority(NBA) for applying for intellectual property rights (including patents) in or outside India needs to be made as per the Biological Diversity Act, 2002.**

No

- iii. Please indicate in case you need assistance to make an application to the NBA.**

Yes.

- iv. Please provide sequence listing in computer readable format.**

- v. In case you would like us to prepare the sequence listing for submission to the Patent Office please indicate**

YES.

- vi. Please indicate if the invention relates to novel biological material for example, bacteria, fungi, eukaryotic cell lines, plant spores, genetic vectors (such as plasmids or bacteriophage vectors or viruses) containing a gene or DNA fragments, or organisms used for expression of a gene (making the protein from the DNA).**

No

- vii. If Yes, have you deposited material with the recognized depositary under the Budapest Treaty?**

(Please note, in case of novel material as mentioned above, deposition must be made before filing of the patent application).

No